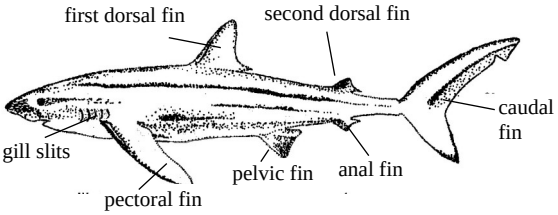
It's all in the key

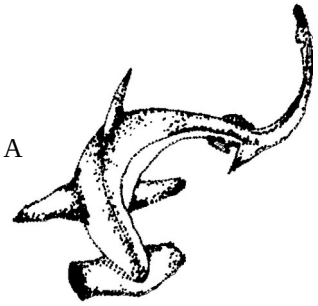
Scientists often classify objects or living things into groups by selecting features or characteristics that some of the things have and some that they don't have. The information that is used to classify these things is sometimes put into a **key**. These keys can then be used to decide which group an object or living thing belongs to or to find the identity of an unknown object or living thing.
The keys used below are called **dichotomous keys** because there are only two choices at each branch.

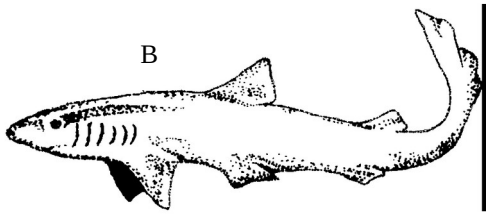
Part A. - Sharks!

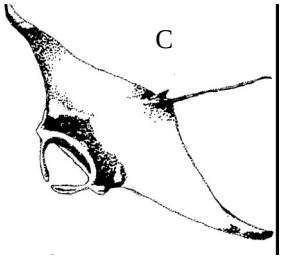
Use the diagram of the shark and the branching dichotomous key below to identify the sharks (A - G) drawn below:

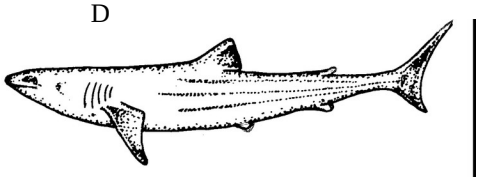


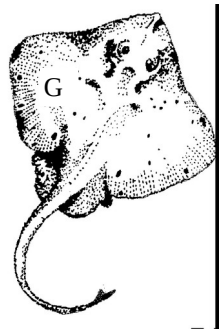
- Fish with skeleton of cartilage
  - Body kite shaped when seen from above
    - Small dorsal fin near tail tip → **True Skate**
    - No dorsal fin
      - Two horn shaped parts on front of animal → **Devil Ray**
      - No horn shaped parts → **Stingray**
  - Body not kite shaped
    - Pelvic fin present
      - Six gill slits → **Six-gilled Shark**
      - Five gill slits
        - Two dorsal fins
          - Mouth on under side of head
            - Head not wider than body
              - Caudal fin not symmetrical
                - First dorsal fin longer than pectoral fin → **False Cat Shark**
                - First dorsal fin shorter than pectoral fin
                  - Caudal fin almost as long as body → **Thresher Shark**
                  - Caudal fin regular length; nose long and pointed → **Goblin Shark**
              - Caudal fin symmetrical → **Mackeral Shark**
            - Head wider than body → **Hammerhead Shark**
          - Mouth at front of snout → **Whale Shark**
        - One dorsal fin → **Cat Shark**
    - No pelvic fin: nose saw shaped → **Saw Shark**



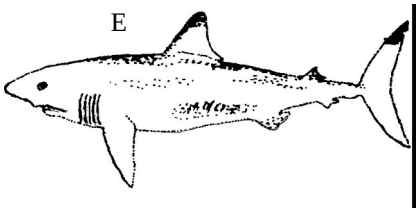




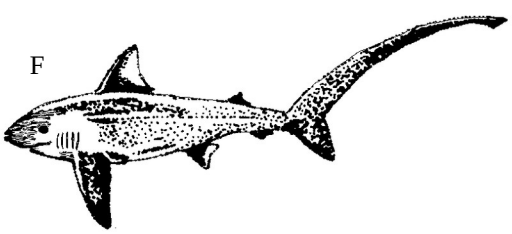




E



F



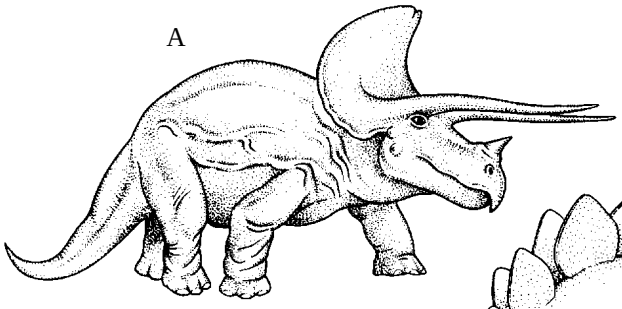
Part B. - Dinosaurs

Use the key below to name the dinosaurs shown:

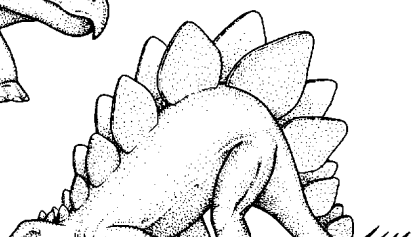
Dinosaur key

1	Limbs used for flying ……...	Go to 2
	Limbs not used for flying …….	Go to 3
2	Long tail ……..	Rhamodactyl (Ram-o-dak-tile)
	Short tail ……..	Pterodactyl (Tero-dak-tile)
3	Limbs used for swimming …….	Go to 4
	Limbs not used for swimming …….	Go to 5
4.	Small head and long neck …….	Plesiosaurs (Please-ee-o-saws)
	Large head and short neck ……	Pliosaurs (Ply-o-saws)
5	All four limbs used for walking ……	Go to 6
	Only hind limbs used for walking ……	Go to 8
6	Long bony frill on head ……	Go to 7
	No frill on head ……..	Stegosaurus (Steg-o-saw-rus)
7	Large horns on head ……..	Chasmosaurus (Chas-mo-saw-rus)
	No horns on head ……..	Protoceratops (Pro-toe-care-a-tops)
8	Bony helmets or crests on head …….	Saurolophus (Saw-ro-lo-fuss)
	No bony projections on head …….	Go to 9
9	Large curved claw on foot ……	Deinonychus (Dine-on-eye-cus)
	No large claw ……..	Tyrannosaurus (Tie-ran-no-saw-rus)

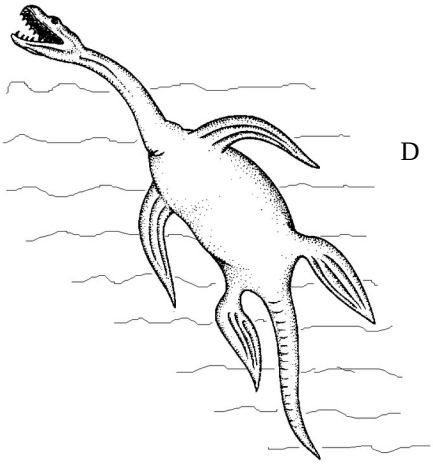
A



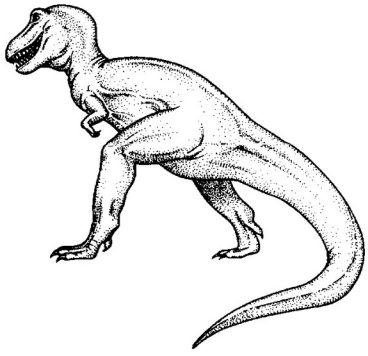
B



C



D



Part C. - Eight Powders

Obtain the eight powders labelled 1 - 8, from your teacher and then use the following sentence key to identify each powder.

Key to the eight powders 1 - 8	
1. Is the powder white?	Yes - go to 2 No - go to 6
2. Is the powder soluble?	Yes - **Glucose** No - go to 3
3. Does the powder bubble when acid is put on it?	Yes - **Chalk** No - go to 4
4. Does iodine turn the powder blue-black?	Yes - **Starch** No - go to 5
5. Does the powder dissolve in acid?	Yes - **Calcium oxide** No - **Aluminium oxide**
6. Does the powder dissolve in acid to form a blue solution?	Yes - **Copper oxide** No - go to 7
7. Is the powder yellow?	Yes - **Sulfur** No - **Charcoal**

Substance 1. ..... Substance 2. .....

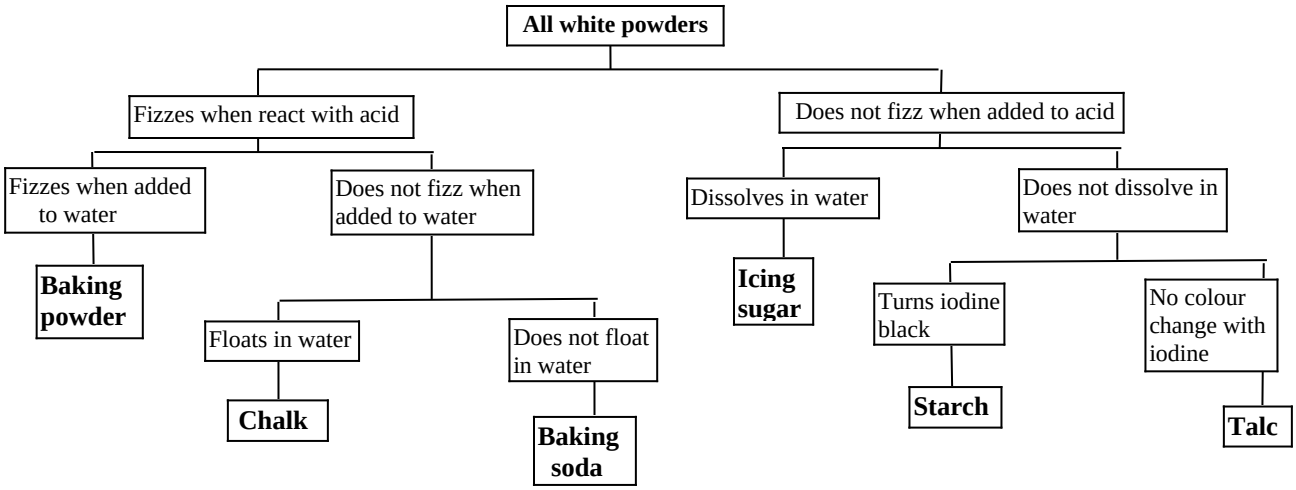
Substance 3. ..... Substance 4. .....

Substance 5. ..... Substance 6. .....

Substance 7. ..... Substance 8. .....

Part D. - White Powders

In this experiment you will use the key given below to identify six unknown white powders, A - F.



Substance A .....

Substance B .....

Substance C .....

Substance D .....

Substance E .....

Substance F .....

Part E - Animals

Construct a branching dichotomous key (similar to the keys shown in Parts A and D) to identify the following animals:

